WIN-IO-8AI-ME

Technical Datasheet Input / Output Modules with Modbus TCP Protocol with Ethernet Interface

The IO module communicates via Ethernet Interface. The module is designed to allow interface with other Modbus TCP/IP compatible devices. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.
Digital IO module is sturdy, low power usage and easy to use.

8 Port AI Module with Modbus TCP Ethernet Port: -

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates ‘**resettable Fuses**’ to safeguard against sudden High Voltage spikes. The board also has ‘**reverse polarity’** protection both for input **Power** connector.

The communication port also has signal isolation and additional Jumper for ‘end of line’ 120 ohms termination resistance which may or may not be used.

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Specifications

General: –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	70 mm L x 70 mm B x 50 mm H
Power	Input Power – 12 – 24 VDC or 24 V AC/DC Typical – 12V DC @ 1Amp
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	

AI Inputs: –

Channels	8
Input Signal	4 – 20 mA / 0 – 20 mA / 0 – 10 V (jumper selectable)
Accuracy	± 2 % of Full scale
Input Resolution	10-bit /12-bit resolution (optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Additional Features: -

Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air EFT
IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
Web based Configuration needed for the IO board

Configuration Settings: -

Communication Speed 10/100 Base-T Ethernet compatible Interface
Function Code AI 0x03 Read Holding Registers
AI Register Address **10 Bit** - 0,1,2,3,4,5,6,7 / **12 Bit** – 8, 9,10,11,12,13,14,15

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	**AI 1 – 10 Bit**	40001	0X03	RS485	16 Bit Unsigned int
2	**AI 2 – 10 Bit**	40002	0X03	RS485	16 Bit Unsigned int
3	**AI 3 – 10 Bit**	40003	0X03	RS485	16 Bit Unsigned int
4	**AI 4 – 10 Bit**	40004	0X03	RS485	16 Bit Unsigned int
5	**AI 5 – 10 Bit**	40005	0X03	RS485	16 Bit Unsigned int
6	**AI 6 – 10 Bit**	40006	0X03	RS485	16 Bit Unsigned int
7	**AI 7 – 10 Bit**	40007	0X03	RS485	16 Bit Unsigned int
8	**AI 8 – 10 Bit**	40008	0X03	RS485	16 Bit Unsigned int
9	**AI – 1 12 Bit**	40009	0x03	RS485	16 Bit Unsigned int
10	**AI – 2 12 Bit**	40010	0x03	RS485	16 Bit Unsigned int
11	**AI – 3 12 Bit**	40011	0x03	RS485	16 Bit Unsigned int
12	**AI – 4 12 Bit**	40012	0x03	RS485	16 Bit Unsigned int
13	**AI – 5 12 Bit**	40013	0x03	RS485	16 Bit Unsigned int
14	**AI – 6 12 Bit**	40014	0x03	RS485	16 Bit Unsigned int
15	**AI – 7 12 Bit**	40015	0x03	RS485	16 Bit Unsigned int
16	**AI – 8 12 Bit**	40016	0x03	RS485	16 Bit Unsigned int

Note: -

For MODBUS communications, a shielded and twisted pair cable is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable Shielded and twisted pair should be used.
Tinned Copper Recommended
Nominal Conductor DCR 14.7 ohm / 1000 ft
Nominal Capacitance 11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss 0.5db / 100ft

Configuration Guide – Ethernet Modbus TCP/IP IO Module:-

1. **Open a web browser** (Google Chrome, Microsoft Edge, Firefox, etc.).
2. In the browser's **address bar**, type the **default IP address:**
   192.168.0.160
3. Press **Enter**. A message will appear saying:
   "Please go to http://192.168.0.160/setup**."**
4. **Copy** this link (http://192.168.0.160/setup) and **paste it** into a new browser tab.
5. Press **Enter** again to open the **Setup / Configuration Page** of the device.

To Reset the Device to Default Settings

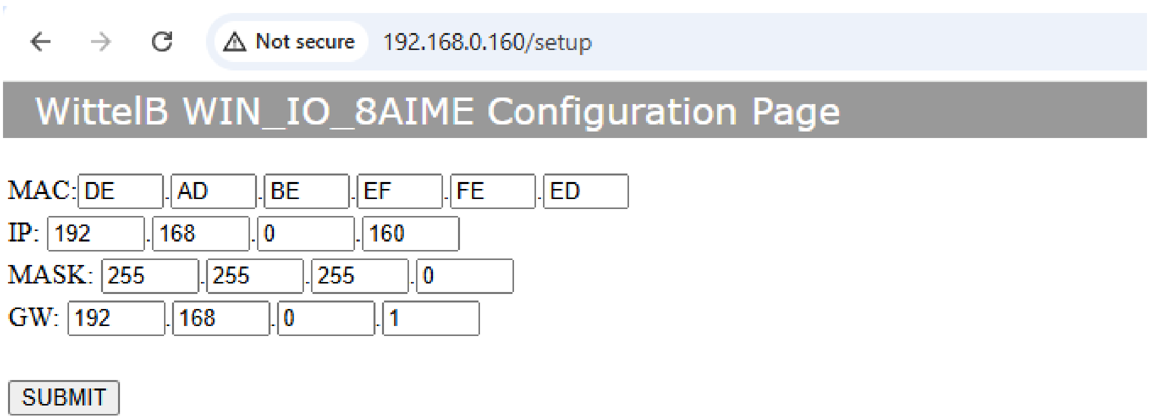
- Place the **jumper** on the designated **Reset position** on the device.
- Press the **Reset button** once.
- The module will reboot and revert to its **default IP address: 192.168.0.160.**

To Change the IP Address

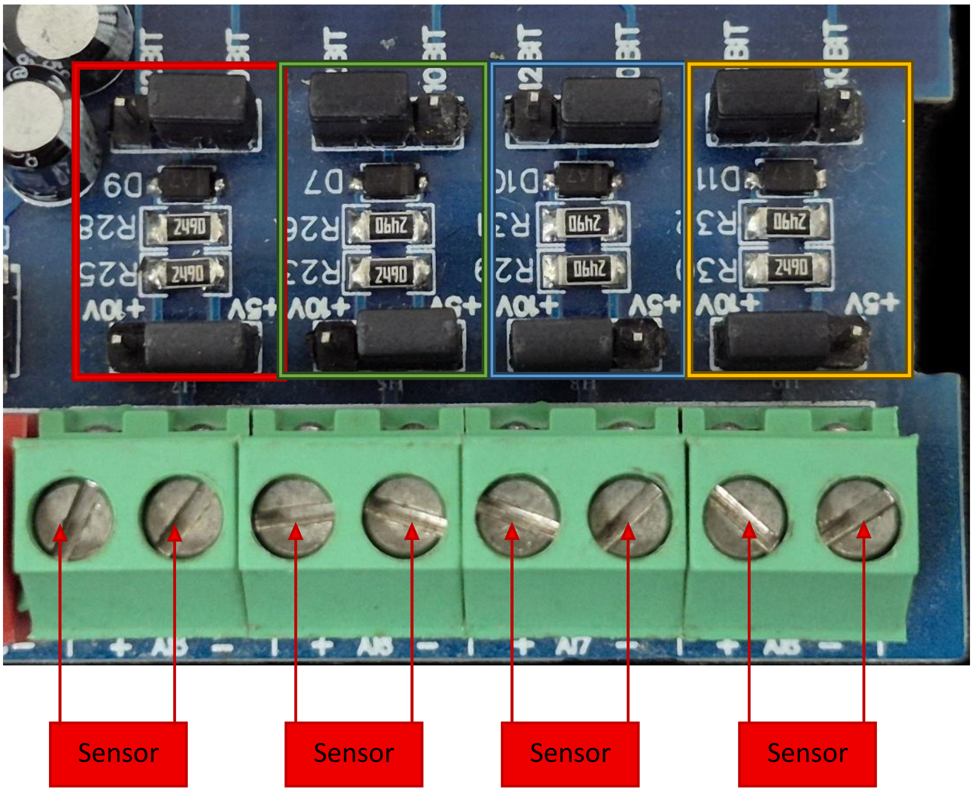
- On the **Setup / Configuration Page**, enter the **new IP address** you wish to assign.
- Click **Submit** to save the new settings.
- The device will automatically apply the **new IP address**.

Note:

- Ensure your PC's IP address is in the **same network range** (for example, **192.168.0.xxx**) when accessing the module.
- After changing the IP, you must use the **new address** to log in again.



AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: **+** (Positive) and **-** (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
  - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
  - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
  - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
  - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
  - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
  - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.

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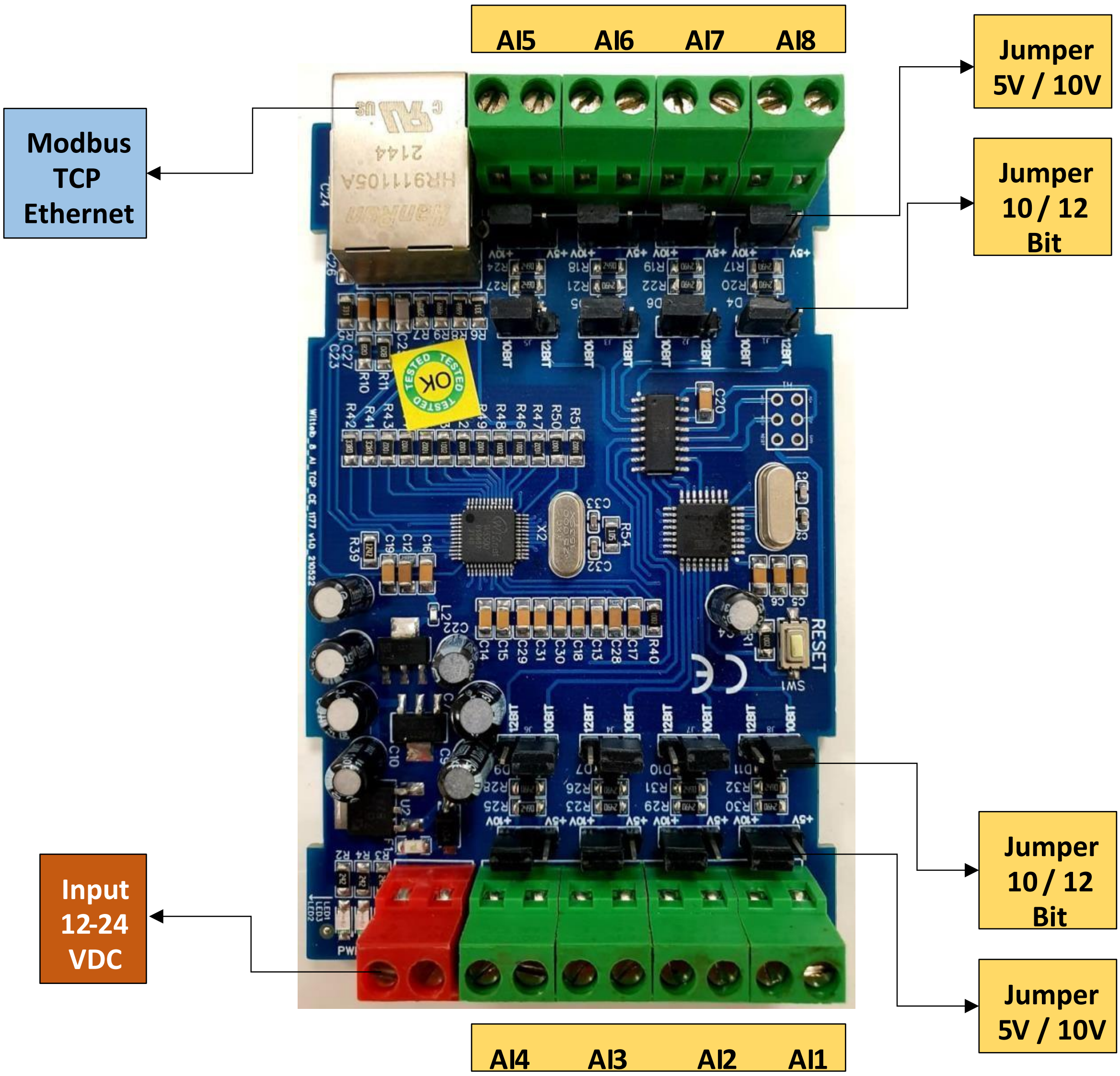

NOTE:–

By default, the Jumper setting on the board is 10 V and 10 Bit.

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